

Git Setup Commands

Check Git version

```
git --version
```

Configure username

```
git config --global user.name "Your Name"
```

Configure email

```
git config --global user.email "you@example.com"
```

Check configuration

```
git config --list
```

Initialize Repository

Create a new Git repository

```
git init
```

Clone Repository

Copy existing GitHub repo to local machine

```
git clone <repo-url>
```

Example:

```
git clone https://github.com/user/project.git
```

Check Status

See changed files

```
git status
```

Add Files

Add single file

```
git add filename
```

Add all files

```
git add .
```

Commit Changes

Save changes

```
git commit -m "Your message"
```

Example:

```
git commit -m "Added login page"
```

View Commit History

Show commits

```
git log
```

Short one-line history

```
git log --oneline
```

Connect Local Repo to GitHub

Add remote repository

```
git remote add origin <repo-url>
```

Check connected remote

```
git remote -v
```

Push Code to GitHub

Push code first time

```
git push -u origin main
```

Push later changes

```
git push
```

Pull Latest Code

Download latest changes

```
git pull
```

Branch Commands

See branches

```
git branch
```

Create new branch

```
git branch branch-name
```

Switch branch

```
git checkout branch-name
```

Create + switch together

```
git checkout -b branch-name
```

Modern alternative

```
git switch branch-name
```

Delete branch

```
git branch -d branch-name
```

Merge Branches

Merge another branch into current branch

```
git merge branch-name
```

Undo Changes

Remove file from staging

```
git restore --staged filename
```

Discard changes in file

```
git restore filename
```

Reset last commit

```
git reset --soft HEAD~1
```

Stash Commands

Temporarily save changes

```
git stash
```

Restore stashed changes

```
git stash pop
```

See stashes

```
git stash list
```

Fetch Commands

Get latest updates without merging

```
git fetch
```

Delete Repository Tracking

Remove remote

```
git remote remove origin
```

Helpful Shortcut Workflow

Most common daily workflow:

```
git status  
git add .  
git commit -m "message"  
git push
```

GitHub Workflow Example

```
git clone repo-url  
git checkout -b feature-login  
git add .  
git commit -m "Created login page"  
git push origin feature-login
```